

Расчетно-графическая работа № 3
Тема: «Теория графов»

Выбор варианта расчетно-графической работы. По строкам располагаются последние цифры номера студенческого билета, по столбцам – предпоследняя цифра номера студенческого билета. На пересечении соответствующих цифр ваш вариант!!!

Варианты РГР № 3		Последняя цифра номера студенческого билета									
		0	1	2	3	4	5	6	7	8	9
Предпоследняя цифра номера студенческого билета	0	1	2	3	4	5	6	7	8	9	10
	1	11	12	13	14	15	16	17	1	2	3
	2	4	5	6	7	8	9	10	11	12	13
	3	14	15	16	17	1	2	3	4	5	6
	4	7	8	9	10	11	12	13	14	15	16
	5	17	1	2	3	4	5	6	7	8	9
	6	10	11	12	13	14	15	16	17	1	2
	7	3	4	5	6	7	8	9	10	11	12
	8	13	14	15	16	17	1	2	3	4	5
	9	6	7	8	9	10	11	12	13	14	15

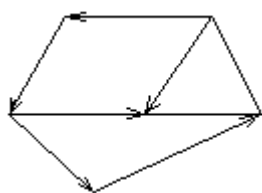
Условия к заданиям для всех вариантов общие:

- В графах G_1 и G_2 пометить вершины и дуги (в графе G_2).
 - Построить матрицу смежностей графа G_1 ;
 - Построить матрицу смежностей и инцидентностей мультиграфа G_2 ;
 - Восстановить граф по матрице смежностей A_G . Задать G с помощью списка дуг и с помощью структуры смежности.
- Даны графы G_1 и G_2 . Построить:
 $G_1 \cap G_2, G_1 \cup G_2, G_1 \oplus G_2, \bar{G}_1, \bar{G}_2, G_1 \times G_2, G_1 [G_2]$. Вершины пометить самим.
- Построить:
 - граф гомоморфный функции;
 - изоморфный функции;
 - граф, являющийся афтоморфизмом данного.
- Найти матрицу достижимости, контрдостижимости. Указать все сильные компоненты связности графа.
- Определить диаметр, радиус и центр графа.
- Пометить вершины. Из неорграфа получить контурный орграф. Расставить веса дуг. Найти кратчайшее расстояние от вершины 1 до всех остальных (вершин).
 - Из неорграфа получить бесконтурный орграф. Найти кратчайшее расстояние от вершины 1 до всех остальных во взвешенном бесконтурном орграфе.
 - Найти один из кратчайших маршрутов (любой).

Вариант 1

1.

a)



G_1

б)

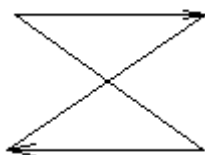


G_2

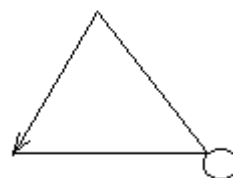
в)

$$A_G = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

2.

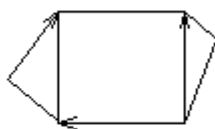


G_1

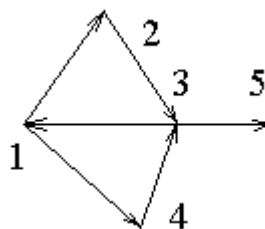


G_2

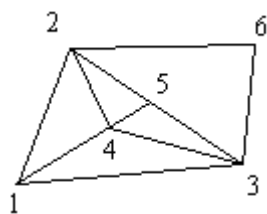
3.



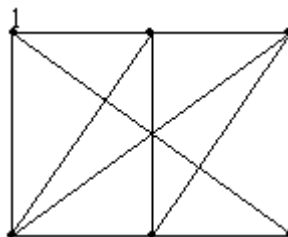
4.



5.



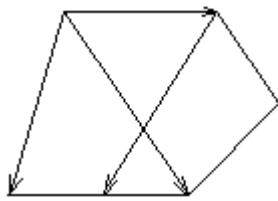
6.



Вариант 2

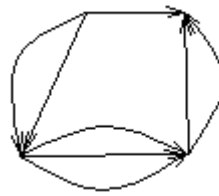
1.

а)



G_1

б)

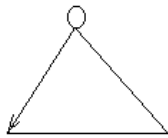


G_2

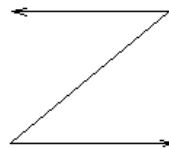
в)

$$A_G = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

2.

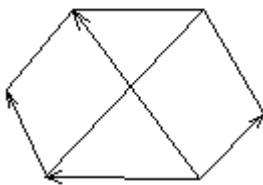


G_1

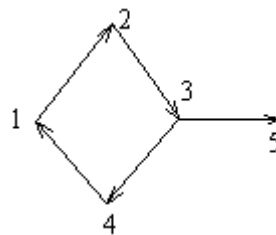


G_2

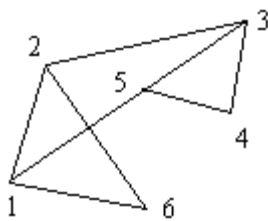
3.



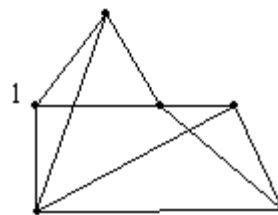
4.



5.

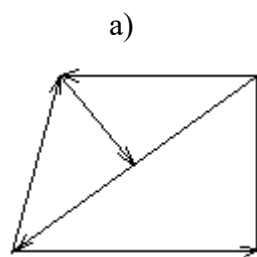


6.



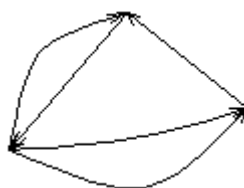
Вариант 3

1.



G_1

б)

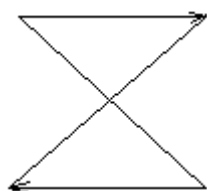


G_2

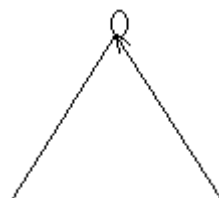
в)

$$A_G = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

2.



G_1

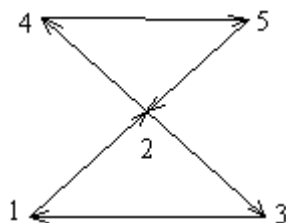


G_2

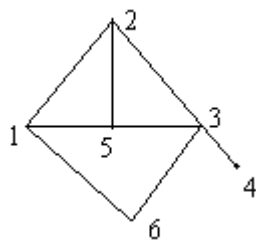
3.



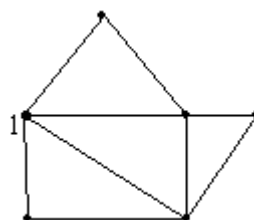
4.



5.



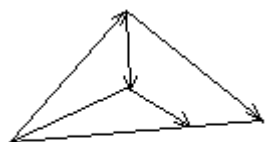
6.



Вариант 4

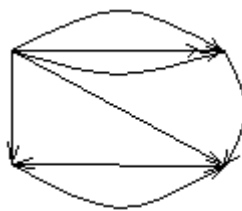
1.

а)



G_1

б)



G_2

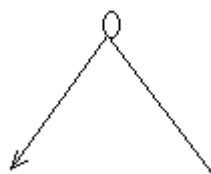
в)

$$A_G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

2.

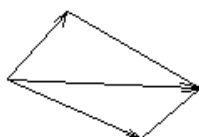


G_1

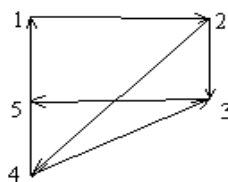


G_2

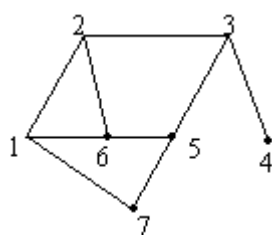
3.



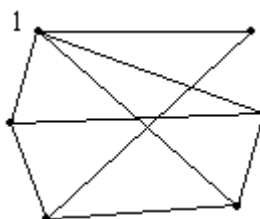
4.



5.



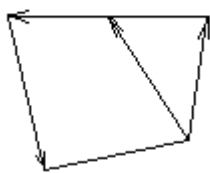
6.



Вариант 5

1.

a)



G_1

б)



G_2

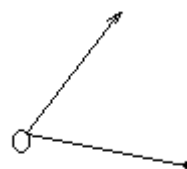
в)

$$A_G = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

2.

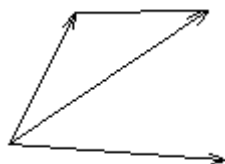


G_1

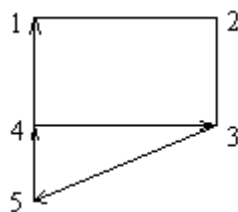


G_2

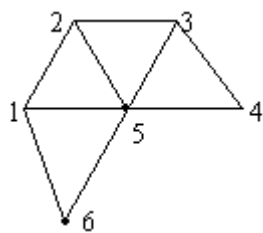
3.



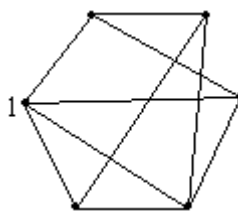
4.



5.



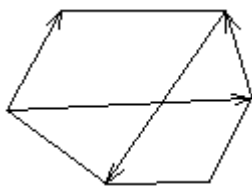
6.



Вариант 6

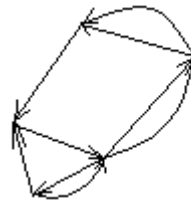
1.

а)



G_1

б)

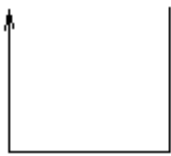


G_2

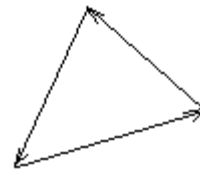
в)

$$A_G = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

2.

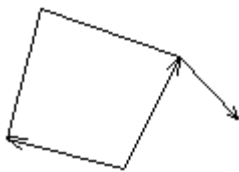


G_1

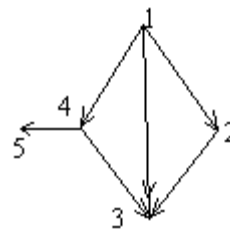


G_2

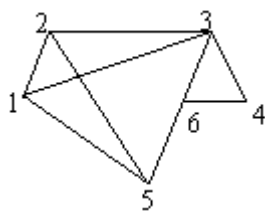
3.



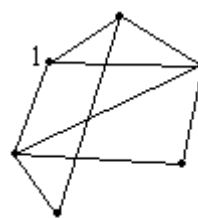
4.



5.



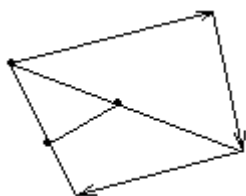
6.



Вариант 7

1.

a)



G_1

б)

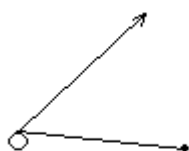


G_2

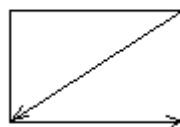
в)

$$A_G = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

2.



G_1

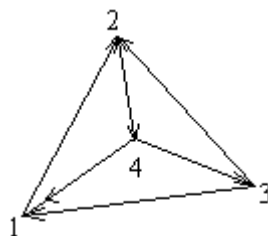


G_2

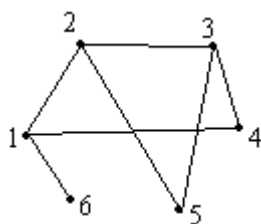
3.



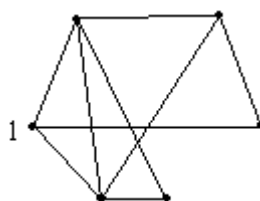
4.



5.



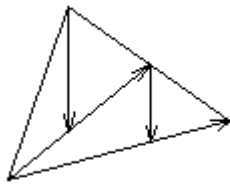
6.



Вариант 8

1.

a)



G_1

б)



G_2

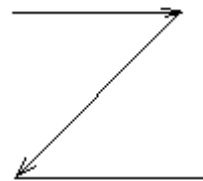
в)

$$A_G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

2.

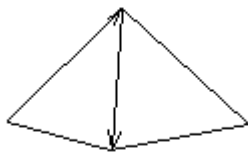


G_1

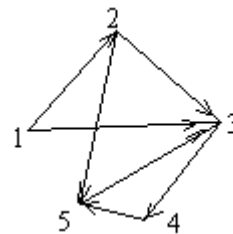


G_2

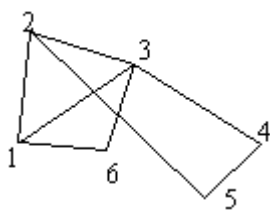
3.



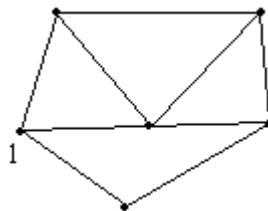
4.



5.

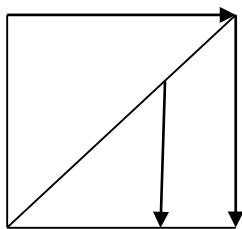


6.



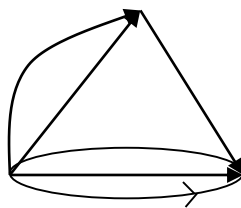
Вариант 9

1. а)



G1

б)

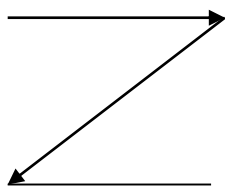


G2

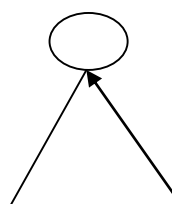
в)

$$A_B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

2.

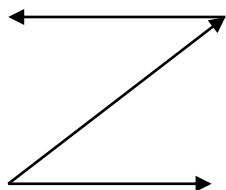


G1

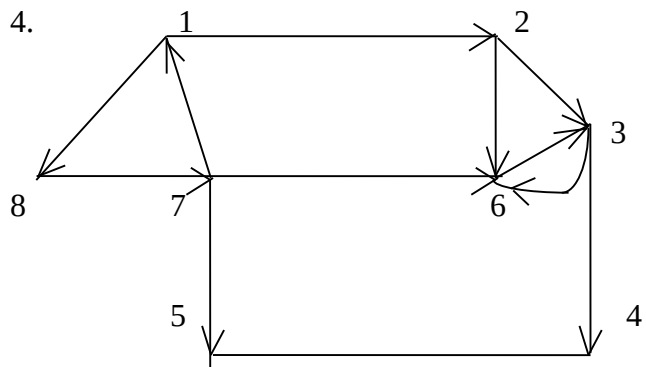


G2

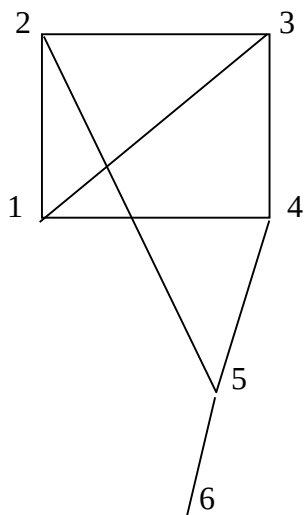
3.

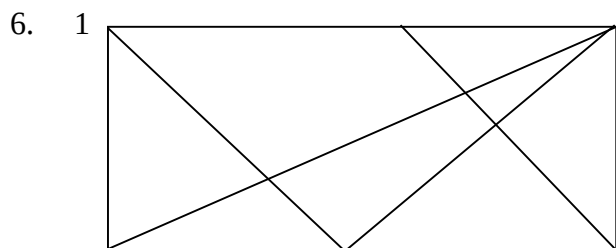


4.



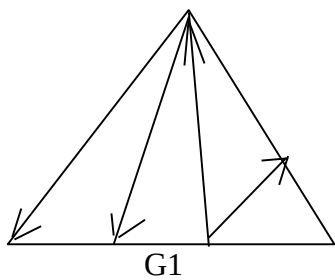
5.



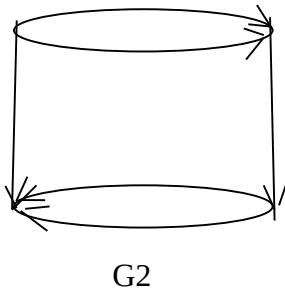


Вариант 10

1. a)



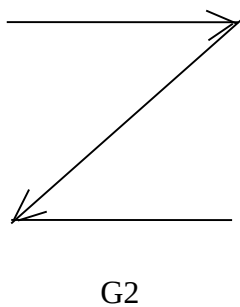
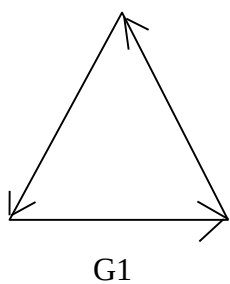
б)



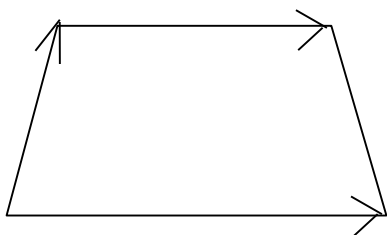
в)

$$A_B = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

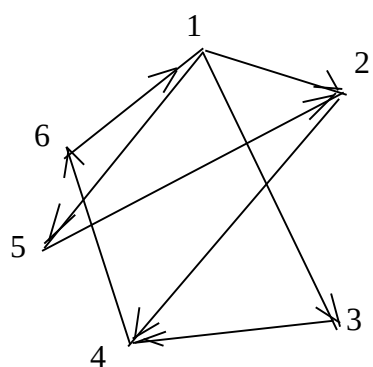
2.



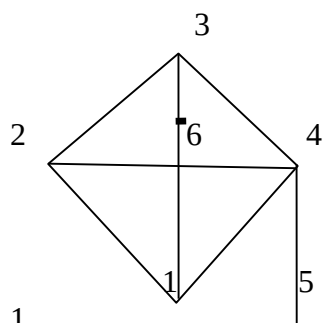
3.



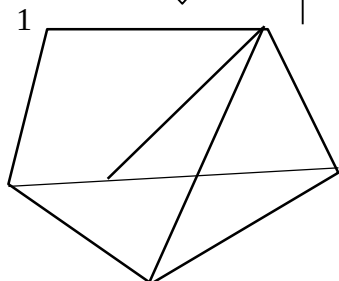
4.



5.

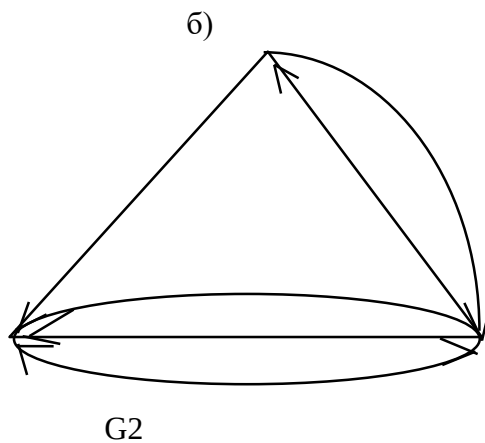
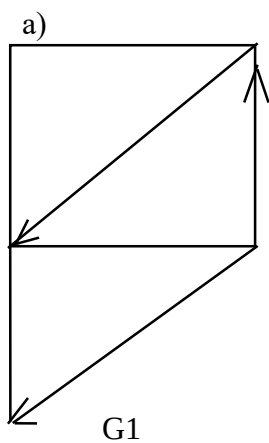


6.



Вариант 11

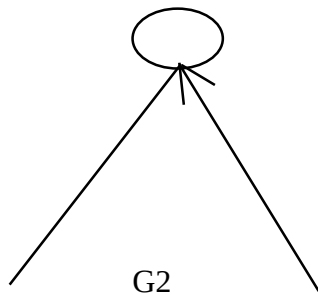
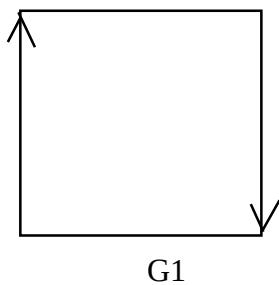
1.



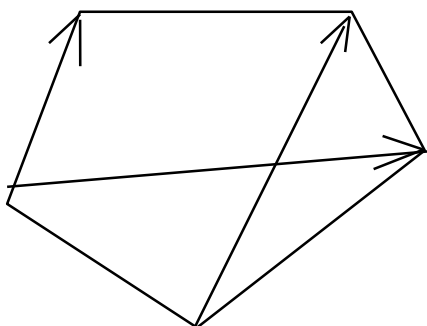
в)

$$A_B = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

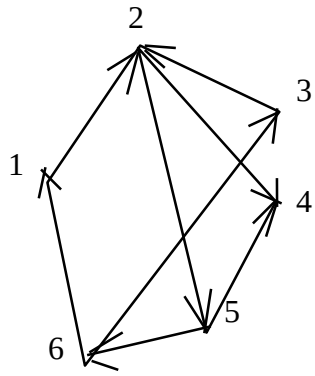
2.



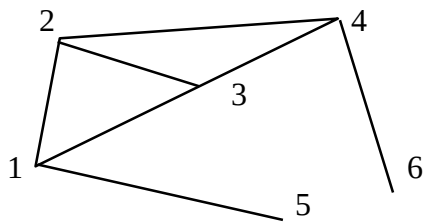
3.



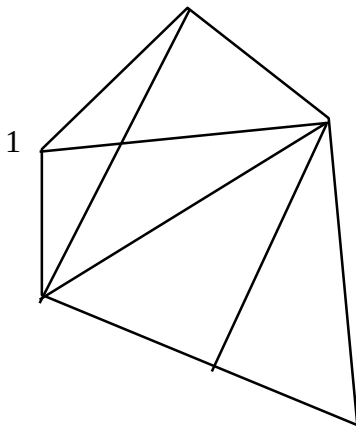
4.



5.



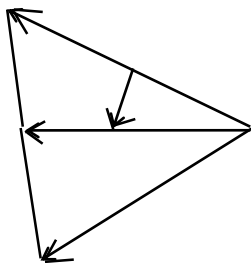
6.



Вариант 12

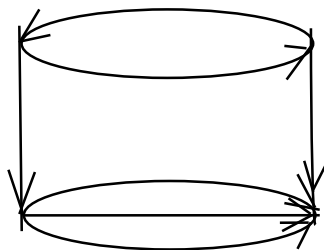
1.

а)



G1

б)

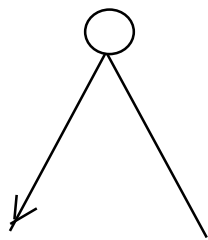


G2

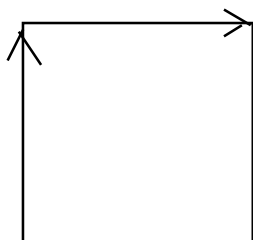
в)

$$A_B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

2.

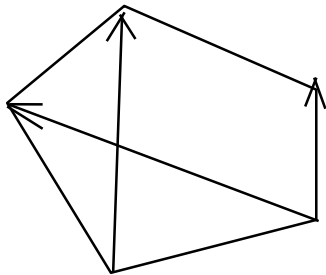


G1

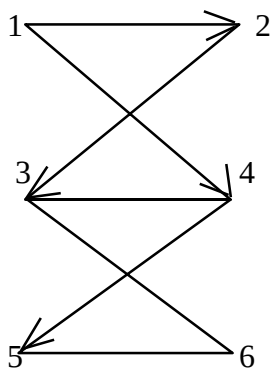


G2

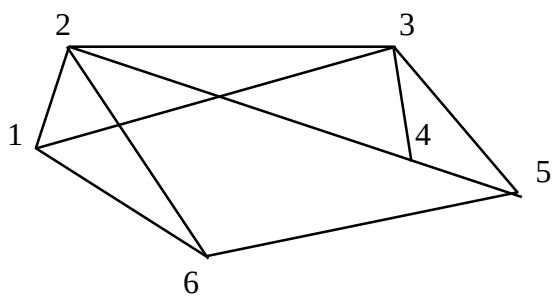
3.



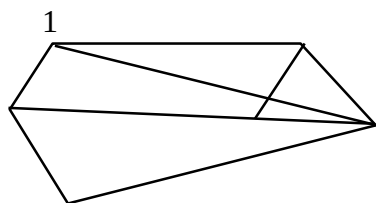
4.



5.



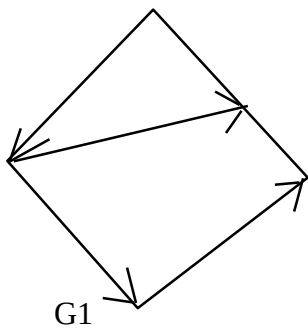
6.



Вариант 13

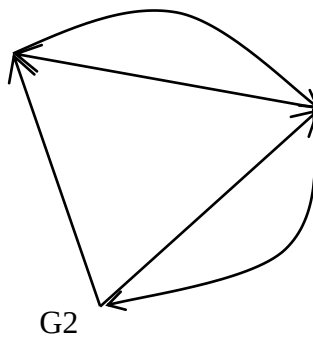
1.

a)



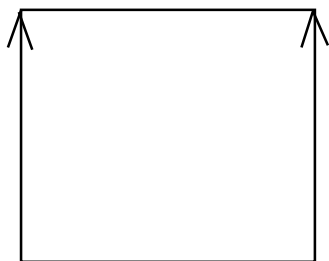
б)

б)

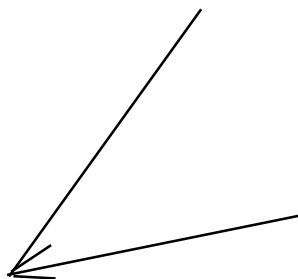


$$AB = \begin{pmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

2.

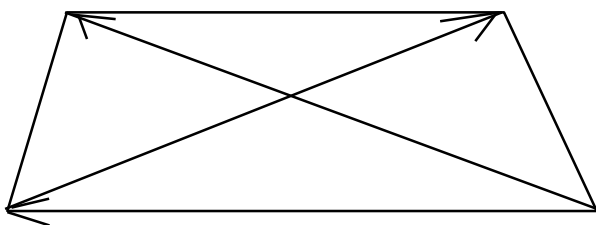


G1

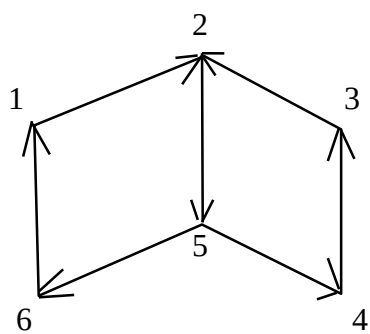


G2

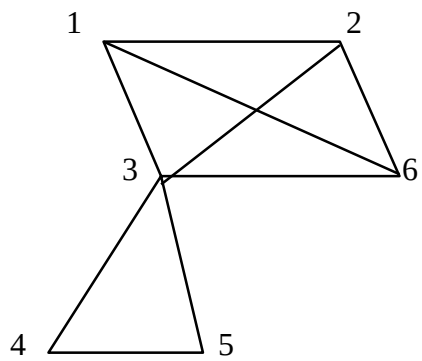
3.



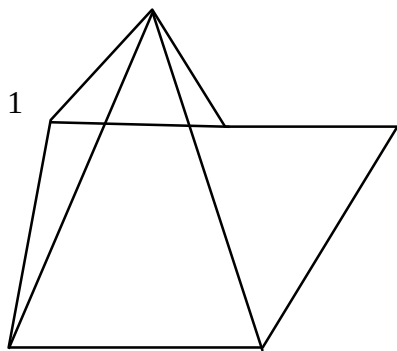
4.



5.



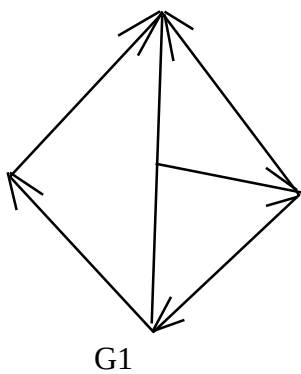
6.



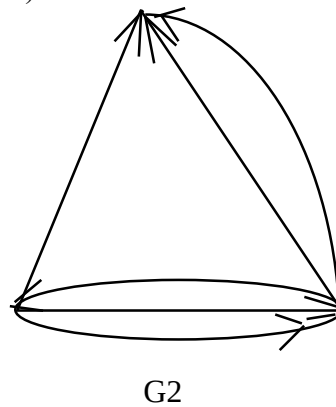
Вариант 14

1.

a)



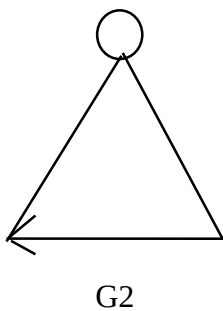
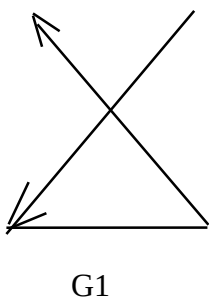
б)



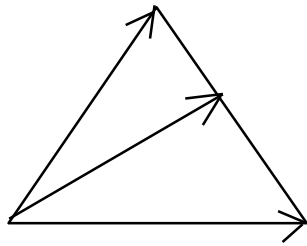
в)

$$A_B = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

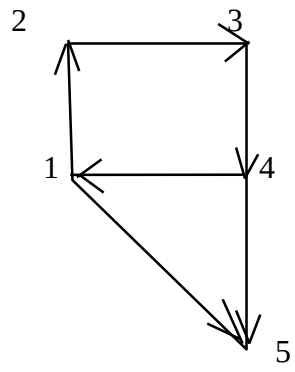
2.



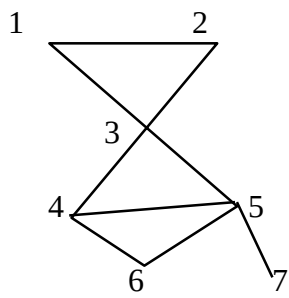
3.



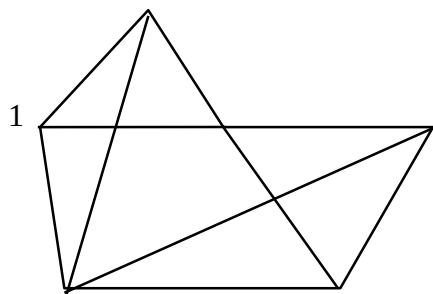
4.



5.



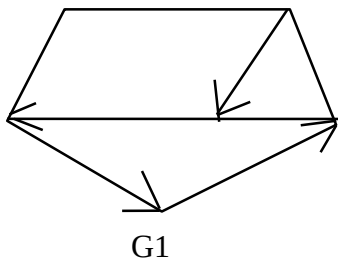
6.



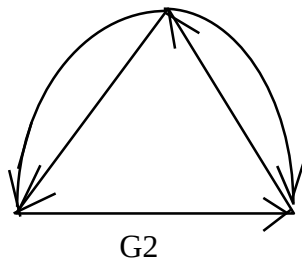
Вариант 15

1.

a)



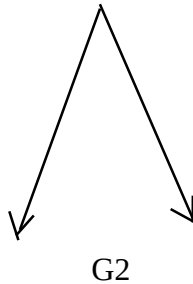
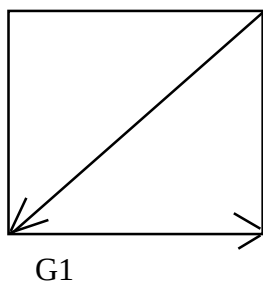
б)



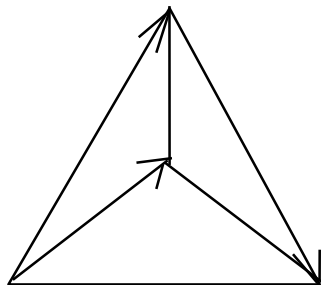
в)

$$A_B = \begin{pmatrix} 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

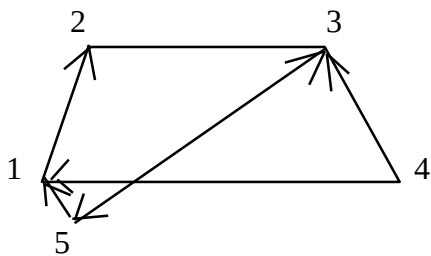
2.



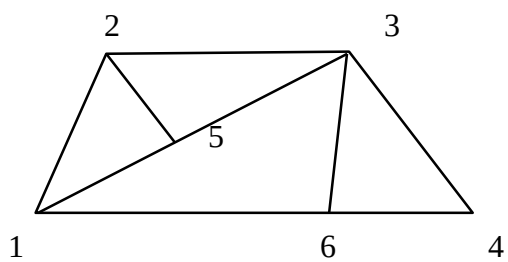
3.



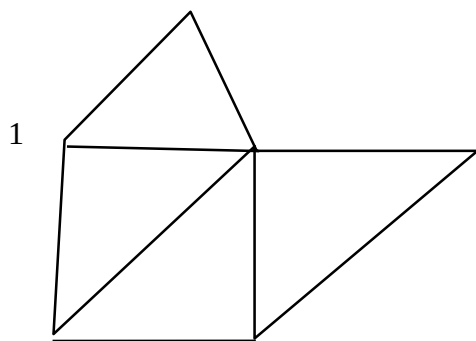
4.



5.



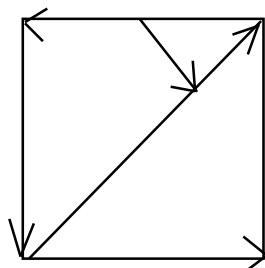
6.



Вариант 16

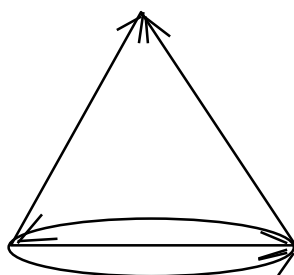
1.

a)



G1

б)

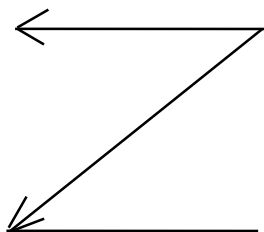


G2

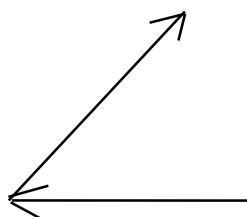
в)

$$AB = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

2.

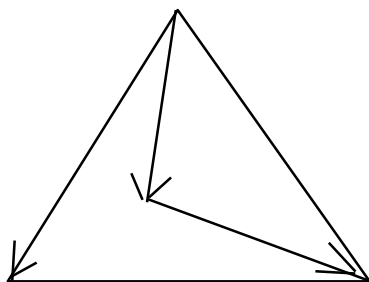


G1

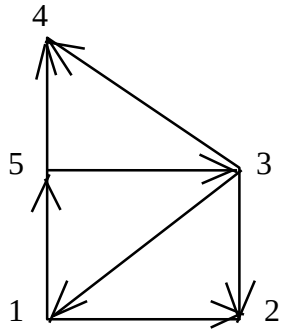


G2

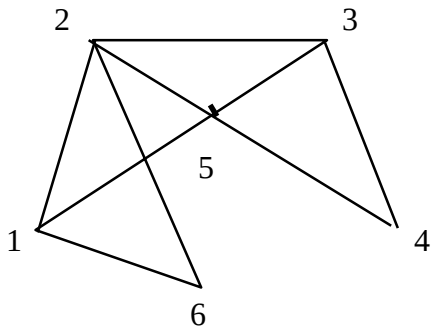
3.



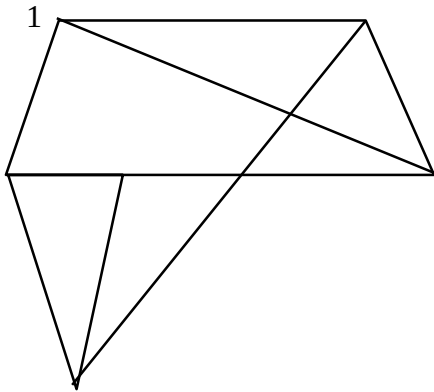
4.



5.



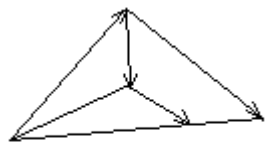
6.



Вариант 17

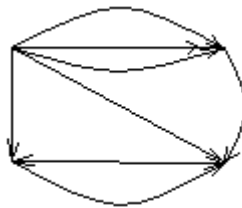
1.

a)



G_1

б)

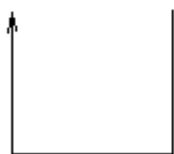


G_2

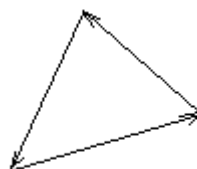
в)

$$A_G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

2.

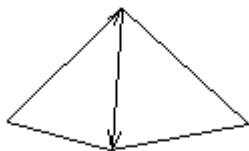


G_1

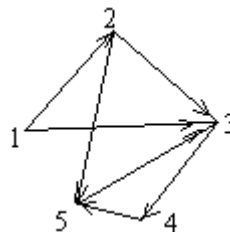


G_2

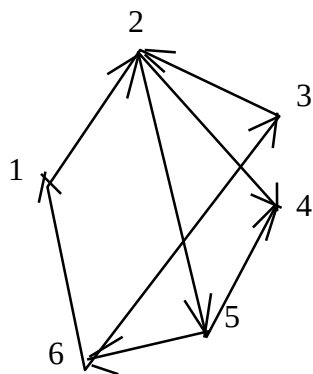
3.



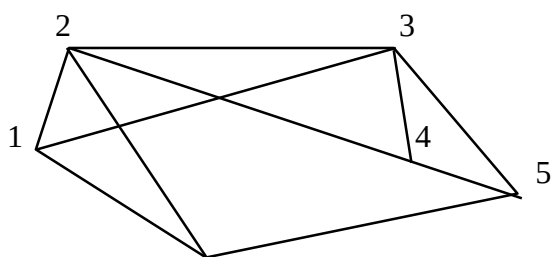
4.



4.



5.



6.

